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56.

SOLUTION :

Given :

n = 1mole

P = 4.1 atm

V=2 Lit

T = ?

q=3710 J

$$\Delta S = \frac{q}{T}$$

$$\Delta S = \frac{q}{\left(\frac{PV}{nR}\right)}$$

$$\Delta S = \frac{nRq}{PV}$$

$$\Delta S = \frac{1 \times 0.082 \text{ lit atm } K^{-1} \times 3710 \text{ J}}{4.1 \text{ atm} \times 2 \text{ lit}}$$

$$\Delta S = \frac{1 \times 0.082 \text{ lit atm } K^{-1} \times 3710 \text{ J}}{4.1 \text{ atm} \times 2 \text{ lit}}$$

$\Delta S = 37.10 \text{ JK}^{-1}$

57.

SOLUTION:

Given :

$\Delta H_f(NaCl) = 30.4 \text{ kJ} = 30400 \text{ J mol}^{-1}$

$\Delta S_f(NaCl) = 28.4 \text{ JK}^{-1} \text{ mol}^{-1}$

$T_f = ?$

$$\Delta S_f = \frac{\Delta H_f}{T_f}$$

$$T_f = \frac{\Delta H_f}{\Delta S_f}$$

$$T_f = \frac{30400 \text{ J mol}^{-1}}{28.4 \text{ J K}^{-1} \text{ mol}^{-1}}$$

$T_f = 1070.4 \text{ K}$

58.

SOLUTION :

Given

$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$

$\Delta H_C^0 = -2220.2 \text{ kJ mol}^{-1}$ -----(1)

$C + O_2 \rightarrow CO_2$

$\Delta H_f^0 = -393.5 \text{ kJ mol}^{-1}$ -----(2)

$H_2 + \frac{1}{2} O_2 \rightarrow H_2O$

$\Delta H_f^0 = -285.8 \text{ kJ mol}^{-1}$ ----- (3)

$3C + 4H_2 \rightarrow C_3H_8$

$\Delta H_C^0 = ?$

$(2) \times 3 \Rightarrow 3C + 3O_2 \rightarrow 3CO_2$

$\Delta H_f^0 = -1180.5 \text{ kJ}$ -----(4)

$(3) \times 4 \Rightarrow 4H_2 + 2O_2 \rightarrow 4H_2O$

$\Delta H_f^0 = -1143.2 \text{ kJ}$ -----(5)

$(4) + (5) - (1) \Rightarrow 3C + 3O_2 + 4H_2 + 2O_2 + 3CO_2 + 4H_2O \rightarrow 3CO_2 + 4H_2O + C_3H_8 + 5O_2$

$\Delta H_f^0 = -1180.5 - 1143.2 - (-2220.2) \text{ kJ}$

$3C + 4H_2 \rightarrow C_3H_8$

$\Delta H_f^0 = -103.5 \text{ kJ}$

Standard heat of formation of propane is $\Delta H_f^0(C_3H_8) = -103.5 \text{ kJ}$